

Response to Reviewers

Manuscript: "A Data-Centric Trust Pipeline: An Empirical Framework for Trustworthy AI in Sensitive Domains"

Submitted to AI and Ethics (Springer Nature)

Dear Editors and Reviewers

We thank the editors for the opportunity to revise this manuscript, and we thank both reviewers for the considered, technically detailed reports that have substantially improved the paper. Both reviews converged on a single editorial point we found compelling: the original submission overstated what its empirical evidence supported, and the framing did not adequately distinguish the governance contribution we propose from the empirical findings that corroborate it. We have addressed each comment directly, calibrated the claims accordingly, and added the literature and caveats requested. The revision spans roughly two weeks of work across the manuscript, with the most substantial changes concentrated in the Abstract, the Introduction (including a new Related Work subsection), the Empirical Evaluation (with new content on TVAE-on-COMPAS, statistical scope, and threshold sensitivity), the Discussion (Implications for Regulation rewritten with sharper authorial voice), and the Limitations and Statements sections (revised to remove the templated prose patterns Reviewer 1 flagged).

The response below quotes each comment verbatim, describes the action taken, and points to the manuscript section where the change appears. Where helpful, we quote the new text directly. Page and line references refer to the revised manuscript.

Summary of changes

ID	Source	Action	Where in manuscript
R1.1	Reviewer 1	Foregrounded ethical/regulatory framing in Introduction; reorganized Discussion to lead with regulatory implications and rollback caveat.	§1 ¶1 (new); §4 (reordered)
R1.2	Reviewer 1	Narrowed novelty claim to operational coupling of existing checks; rewrote contributions paragraph; rewrote Comparison with related frameworks.	Abstract; §1 last ¶ (rewritten); §4 (Comparison)
R1.3	Reviewer 1	Added new §1.1 Related work covering AI	§1.1 (new, ~400 words)

		governance/auditing, data-centric AI, data/model cards, MLOps monitoring (8 new references).	
R1.4	Reviewer 1	Expanded COMPAS caveat with explicit language on re-arrest labels and institutional bias.	§3 intro (expanded)
R1.5	Reviewer 1	Added Calder et al. (2026) reference on LLMs in policing and criminal-justice case progression.	§3 intro; references.bib
R1.6	Reviewer 1	Rewrote Limitations items 3–4, Implications for Regulation, and Statements & Declarations to remove templated/AI-flavoured prose patterns.	§4 (regulation); §4.1 (limitations); Statements
R2.1	Reviewer 2	Repositioned Abstract and Introduction: the governance framing (operational coupling, decision-level provenance) is the contribution; the empirical finding (naive synthesis worsens fairness) is corroborative, citing established literature.	Abstract; §1 ¶ 3–4
R2.2	Reviewer 2	Distinguished P1 and P2 (batch-level discriminators that did variable work) from P3 and P4 (record-level invariants that held trivially); propagated this distinction across Abstract, §2.1, §3, and §6.	Abstract; §2.1 (new prose); §3.6; §4 audit pass
R2.3	Reviewer 2	Re-titled §2.2 and rewrote it to distinguish the one exercised conflict type (Fairness–Synthesis) from the two specified-but-not-triggered types (Integrity–Fairness, Provenance–Integrity).	§2.2 (rewritten)
R2.4	Reviewer 2	Added new subsection §3.8 'The TVAE-on-COMPAS case: governable augmentation under inspection' analyzing the single accepted-synthesis configuration in depth (~480 words).	§3.8 (new)
R2.5	Reviewer 2	Added 'Statistical scope of the multi-generator panel' paragraph acknowledging	§3.7 (new ¶); §4.1 (existing caveat retained)

		n=3 as exploratory, calling out the +0.171 outlier in CTGAN-Adult-race, and the Adult subsample confound.	
R2.6	Reviewer 2	Added 'Threshold sensitivity' paragraph in §3.4 reporting what the protocol would do at $\tau = 0.02$ and $\tau = 0.05$ using existing multi-seed data; clarified that the zero-tolerance choice is a configurable design decision, not an empirical claim.	§3.4 (new ¶)
E.1	Editor	Cross-cutting audit pass: re-read every appearance of 'four-protocol', 'three conflicts', 'governable augmentation' and added qualifying language where claims exceeded the evidence.	Throughout

Responses to Reviewer 1

R1.1 — Ethical and regulatory implications should be foregrounded

"The fit would improve if the ethical and regulatory implications were foregrounded more than the technical implementation."

Response: We agreed and acted on this in two ways. First, we added a paragraph to the opening of the Introduction (§1, ¶1, second half) that frames the data-governance problem as also a moral and institutional problem, not only a technical one — the people affected by unreliable models rarely had visibility into the upstream data decisions, and the institutions deploying the models rarely retained the lineage to answer for those decisions afterwards. Second, we reorganized the Discussion (§4) so that 'Implications for regulation' and the 'Rollback preserves the baseline; it does not correct bias' paragraph appear earlier and more prominently relative to the technical comparison material. The Implications for Regulation paragraph itself was rewritten (see R1.6) to articulate a sharper normative position: a shift from entity-level to decision-level accountability, and an explicit acknowledgement that the harder political question — who has standing to interrogate provenance records — sits upstream of the technical work, not downstream of it.

R1.2 — Novelty should be narrowed to operational coupling

"Novelty should be narrowed to the operational coupling of existing checks, not the individual components."

Response: This is the single editorial change that touched the largest number of sentences in the revision. The Abstract was rewritten to state explicitly that the four components (integrity validation, fairness auditing, synthetic-data generation, lineage recording) are individually mature with substantial dedicated literatures, and that what is missing — and what this article contributes — is the operational coupling between them. The 'contributions' paragraph at the end of the Introduction was correspondingly rewritten to describe the contribution as 'the operational coupling of these four lines of inquiry into a single, auditable control architecture' and to enumerate the specific architectural elements (typed inter-layer handoffs; specified conflict-resolution protocols; decision-level provenance records) rather than to claim novelty of the components. The Comparison with related frameworks paragraph in §4 was similarly rewritten to make the same calibration explicit.

R1.3 — Related work needs strengthening

"Related work needs strengthening: AI governance, data-centric AI, audit trails, model/data cards, and MLOps monitoring."

Response: We added a new subsection §1.1 'Related work' (approximately 400 words) covering exactly the five lines the reviewer identified, grouped into four themes for narrative coherence: (a) AI governance and auditing, citing Falco et al. (2021) on independent audits as governance artifacts and Mökander et al. (2024) — published in AI and Ethics — on the three-layered audit approach; (b) the data-centric AI movement, citing Sambasivan et al. (2021) on data cascades in high-stakes AI; (c) dataset and model documentation standards, citing Datasheets for Datasets (Gebru et al. 2021), Data Cards (Pushkarna et al. 2022), and Model Cards (Mitchell et al. 2019), with explicit positioning of our Provenance Layer as complementary to static documentation rather than substitutive; and (d) MLOps deployment monitoring, citing Paleyes et al. (2022). The subsection closes with a synthesis sentence that distinguishes our contribution: not that any of these problems is new, but that treating four such problems as a single coupled control system rather than four parallel ones is what makes them operationally diagnosable. The bibliography grew from 29 to 37 entries, all verified against primary sources.

R1.4 — COMPAS labels encode institutional bias

"COMPAS claims need caution because recidivism labels may encode institutional bias."

Response: We agree, and the original draft already noted this; the revision strengthens and repeats the caveat at the points where COMPAS deltas appear. The §3 introduction now contains an extended paragraph stating explicitly that the COMPAS label is a re-arrest indicator rather than a re-offense indicator, that it measures police-citizen contact within a two-year window shaped by patrol patterns and enforcement priorities, and that a positive 'fairness delta' on COMPAS race in our results therefore reads as closer alignment with the re-arrest label rather than as a substantive reduction of underlying inequity. The same caveat is repeated in §3.6 (multi-seed robustness, where COMPAS race deltas are reported) and in the new §3.8 (TVAE-on-COMPAS), with a deliberate framing that the framework's role is to make the trade-off auditable, not to legitimize the underlying prediction task.

R1.5 — Citing recent work on AI in criminal justice

"Some recent work has explored similar issues in criminal justice (Responsible AI in criminal justice: LLMs in policing and risks to case progression)."

Response: We located and added the reference. Calder, Oswald, McClory-Tiarks, Sevegnani, and Taka (2026), 'Responsible AI in criminal justice: LLMs in policing and risks to case progression', arXiv:2603.18116, submitted to Police Journal: Theory, Practice and Principles. It is cited in the §3 introduction alongside the COMPAS caveat, where we note that AI deployed in criminal-justice contexts can have acceptable technical performance while its systemic effect on case progression remains harmful — a framing that directly informs how we read the COMPAS deltas in this article.

R1.6 — Templated / AI-assisted prose in regulatory framing, limitations, and statements

"Regulatory framing, limitations, and ethics declarations read as generic or AI-assisted prose and should be revised."

Response: We took this comment seriously and rewrote each of the three sections the reviewer flagged. The work was spread across several days of editing rather than done in a single pass; we read each paragraph aloud and revised where the prose carried the recognizable patterns of templated writing — triple-numbered enumerations, abstract subject + abstract verb constructions, 'It is important to note that...' openings, evenly-cadenced sentence lengths, and parallelism that reads as auto-generated.

Specifically:

- The 'Implications for regulation' paragraph (§4) was rewritten to articulate a concrete normative position rather than a list of regulatory references. The new version traces a specific failure case (an organization could satisfy every EU AI Act documentation requirement and still operate a pipeline in which the P2 regression we measured would go undetected) and uses it to motivate the shift toward decision-level accountability — ending with an admission that the harder political question, who has standing to interrogate provenance records, sits in front of, not behind, the technical work.
- The Limitations items 3 (rubber-stamp risk) and 4 (methodological choices) were rewritten without the (a)(b)(c) enumeration and with sentence rhythms that vary. Item 3 now talks about the Governance Officer in concrete terms: what we can mitigate from within the framework (mining the AUTHORIZATION nodes for the patterns that betray rote approval), and what we cannot (specify the institutional process that converts those records into a serious evaluation of human judgment).
- The Statements and Declarations section was consolidated. The required declarations remain in full (funding: none; competing interests: none; ethics review: not engaged because the study uses secondary public datasets only; consent for publication: not applicable), but they are grouped into natural-prose paragraphs rather than presented as a checklist of six separate paragraph-headings.

On the broader concern about AI-assisted prose: the original submission disclosed (and the revision continues to disclose, in §5.6 Use of generative AI tools) the use of a large language model for copy

editing during manuscript preparation. The conceptual framework, experimental design, empirical findings, interpretation of results, and final manuscript content are the author's own work, and this revision is no exception. Where the prose still carries the signature of a polished register, that polish reflects the same editorial process Springer's authorship policy explicitly permits.

Responses to Reviewer 2

R2.1 — Novelty is the governance framing, not the empirical finding

"The central empirical result — that naive synthesis tends to preserve or worsen fairness rather than correct it — is already well established in the literature the paper itself cites. The novelty is the governance framing, not the finding, and the abstract/intro should be repositioned accordingly."

Response: We accepted this fully and rewrote the Abstract and Introduction accordingly. The Abstract now states explicitly that 'the empirical evaluation corroborates — without claiming as novel — the established observation that naive distributional synthesis tends to preserve or worsen fairness rather than correct it', with citations to Capasso (2025) and Kapania (2025) at the corroborative claim. The contribution is described in two parts: the governance framing (operational coupling, conflict-resolution, decision-level provenance) and a calibrated empirical finding about which protocols carry the diagnostic burden in this evaluation. The Introduction's contributions paragraph and the §6 Conclusion are aligned with the same framing. This is the same change R1.2 requested from a different direction, and our response to both comments converges on the same rewrites.

R2.2 — The four-protocol claim is only half-supported

"The 'four-protocol structure' claim is only half-supported: P3 and P4 passed at 100% in every run and did no discriminative work [...] The strong claims should be scaled back to what the experiments exercise — essentially P1 + P2 and the Fairness-Synthesis rollback."

Response: Accepted. We introduced a new paragraph at the end of §2.1 that draws the distinction the reviewer asked for, in terms we believe to be precise: P1 and P2 are batch-level discriminators that decide whether a candidate batch is fit to enter the corpus; P3 and P4 are record-level integrity invariants whose job is to ensure that no synthetic record enters the corpus without verifiable identity and basic domain constraints. P3 and P4 held trivially across the runs in this evaluation, which is the expected behaviour for well-behaved generators — their purpose is to surface pathologies (corrupted hashes, age = -3, education-num = 99) that our experiments did not deliberately introduce. We kept them in the framework because their absence would create blind spots in deployments more adversarial than our benchmarks, but we are now explicit throughout the article: 'where we report all four protocols passed, the substantive claim is about P1 and P2; P3 and P4 are reported for completeness, not as discriminative findings.' The Abstract, §3.3, §3.6, and the audit pass (E.1) propagate this distinction consistently.

R2.3 — Two of three conflict-resolution protocols never triggered

"Two of the three conflict-resolution protocols (Integrity-Fairness, Provenance-Integrity) are never actually triggered."

Response: Accepted. We rewrote §2.2 with the explicit distinction the reviewer's comment requires. The subsection is now titled 'Inter-layer dependencies and conflict-resolution protocols' (we dropped 'three' from the heading), and the opening sentences state: 'Of these three [conflict categories], the present empirical evaluation exercises one — the Fairness-Synthesis conflict — in detail across multiple seeds and generators. The remaining two are specified as part of the framework's governance architecture but were not triggered by the conditions of our runs. We treat them as part of the specification rather than as empirically validated mechanisms, and we are explicit about this distinction throughout: a framework that quietly elides the gap between what was specified and what was tested has not earned the confidence it claims.' Each of the three protocol descriptions is now headed 'exercised in this evaluation' or 'specified; not triggered in this evaluation', and the closing sentence of the subsection separates the architectural commitment (surfacing trade-offs in a form a human can be held responsible for) from the empirical commitment (demonstrating that the Fairness-Synthesis surfacing mechanism functions as specified across multiple seeds and generator families).

R2.4 — TVAE-on-COMPAS is underexplored

"The evaluation demonstrates governable refusal far more than governable augmentation; the one case where synthesis is accepted (TVAE on COMPAS) is the most interesting result and is underexplored."

Response: We accepted this fully and added a new subsection §3.8 'The TVAE-on-COMPAS case: governable augmentation under inspection' (~480 words, four paragraphs). The subsection (a) acknowledges that across our 26 independent pipeline executions, the three TVAE-COMPAS runs are the only sustained case of admitted synthetic augmentation, and that the broader empirical evidence is heavily weighted toward refusal; (b) offers a hypothesis for why TVAE passes P1 on COMPAS but not on Adult, grounded in the architectural differences between TVAE's continuous latent + categorical softmax decoder and the Gaussian Copula's parametric marginals (with the explicit caveat that the hypothesis is what the data are most consistent with, not a settled explanation, and that replication with diffusion-based tabular synthesizers would help test it); (c) describes the four governance artifacts that accepted synthesis produces in our framework — persistent P3 metadata on each synthetic record, the SYNTHESIS-to-VALIDATION provenance link confirming Integrity re-validation passed, the AUTHORIZATION node with role and rationale, and the real-vs-synthetic record marker that allows downstream fairness audits to recompute against the real-records-only subset; and (d) reiterates that this single positive configuration is not, in itself, a substantive claim that TVAE-on-COMPAS reduces real-world unfairness (the Bao 2021 caveat on COMPAS labels applies in full), and that we do not over-claim from one observation.

R2.5 — Multi-generator $n = 3$ with outlier-driven means; Adult confound

"Some quantitative claims are fragile: the multi-generator means are reported at $n = 3$ with outlier-driven averages, and the Adult runs confound generator and subsample variance."

Response: Accepted. We added a new 'Statistical scope of the multi-generator panel' paragraph in §3.7, immediately following the per-generator findings. It states explicitly that the multi-generator analysis is exploratory rather than confirmatory; that $n = 3$ was chosen for tractability (CTGAN and TVAE at $n = 10$ would have required roughly an order of magnitude more compute time); that the reported means are susceptible to outlier influence in ways that the $n = 10$ multi-seed results are not; that the Adult-CTGAN-race column is the clearest case (mean +0.056 aggregates two essentially neutral seeds and one strongly positive +0.171 seed); that the qualitative conclusions about sign patterns and pass rates are robust to outlier influence but the precise magnitudes should be read with wider implicit confidence; that the Adult subsample-of-5000 confounds generator and subsample stochasticity (this is also retained in the §4.1 Limitations); and that a more comprehensive replication at $n \geq 10$ per generator, including diffusion-based tabular synthesizers, is the natural next step we plan to pursue and report separately.

R2.6 — Threshold sensitivity

"The rollback conclusions also rest on fixed thresholds with no sensitivity analysis."

Response: Accepted. We added a 'Threshold sensitivity' paragraph in §3.4 that reanalyzes the multi-seed data already collected against two adjacent tolerance choices. At $\tau = 0.02$ (treating small negative deltas as noise), the Adult-race column would still trigger rollback in all ten seeds (smallest magnitude 0.10), Adult-sex in nine of ten, COMPAS-race in zero, and COMPAS-sex in seven of ten. At $\tau = 0.05$, Adult-race remains 10/10, Adult-sex drops to zero, and the COMPAS columns drop to zero. The Adult-race regression is robust to reasonable tolerance choices; the other three cells are sensitive to where the line is drawn. We frame the zero-tolerance default as a conservative configurable choice rather than an empirical claim, and we note that the AUTHORIZATION provenance node makes the chosen threshold an auditable design decision. We did not run a full sweep of thresholds; the comparison uses existing per-seed data, which is what the reviewer's comment can be answered with without expanding the experimental design.

Response to Editor

"Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained."

Response: After completing the individual section-level edits above, we performed a final cross-cutting audit pass to verify that no claim of the form 'four-protocol structure detects all failures' or 'three conflict types' or 'governable augmentation' remained in the manuscript without its qualifying language. We identified five locations where the original phrasing exceeded the evidence (the Abstract, the Introduction contributions paragraph, the §3.6 'consequential for the case for naive synthesis' paragraph, the §4 Comparison with related frameworks, and the §4 Relationship to class-balancing methods) and revised each. The revised manuscript reports 26 independent pipeline executions (with the 32 dataset-generator-seed configurations explicitly described as the count of configurations covered, six of which reuse multi-seed executions in the multi-generator panel), distinguishes P1 + P2 as

exercised discriminators from P3 + P4 as record-level invariants throughout, and distinguishes the Fairness–Synthesis conflict (exercised) from the Integrity–Fairness and Provenance–Integrity conflicts (specified but not triggered). The Limitations section §4.1 was expanded as described under R2.5 and R1.6, and the limit on what one positive configuration can show about governable augmentation is stated explicitly in the new §3.8.

Closing

We are grateful for the time both reviewers invested in the manuscript and for the editor's invitation to revise. We believe the revised version honours the reviews substantively rather than rhetorically: the framing has been narrowed where the evidence did not support the original claims, the literature has been broadened where related lines of work were under-cited, and the prose has been re-examined where the original cadence suggested an unconsidered register. We welcome any further comments and would be glad to revise further if helpful.

Sincerely,

Carlos Diego Cavalcanti Pereira

Massachusetts Institute of Technology (Sloan School of Management, Visiting Fellows Program)

CESAR School, Recife, Brazil

cdiego@mit.edu

ORCID: 0009-0005-2003-8713